UNIVERSITY OF SOUTH CAROLINA
Electrical Engineering

MID-TERM EXAMINATION FOR

(Spring Semester: 2022)

ELCT 361– ELECTROMAGNETICS

April 8th, 2021 – Time Allowed: 1 hour 30 minutes (10:30 AM ~ 12:00 PM)

INSTRUCTIONS TO CANDIDATES

1. This paper contains **FOUR (4)** questions and comprises **TEN (10)** printed pages.
2. Answer for all problems.
3. Each question carry different **POINTS**.
4. Please write your answers directly in **EXAM PAPER**
5. If you make any assumptions in addition to those stated in the question, please state them explicitly and clearly.

<Solutions>

Student ID:

Name:

Q1 **[25 points]** A plastic sheet with $\boldsymbol{\epsilon_r = 2.8}$ is introduced perpendicularly in a uniform electric field intensity $\overrightarrow{\boldsymbol{E_0}} = \widehat{\boldsymbol{a_x}}\boldsymbol{\epsilon_0 E_0}$ as shown below, (where $\boldsymbol{\epsilon_0}$ is permittivity in free space)

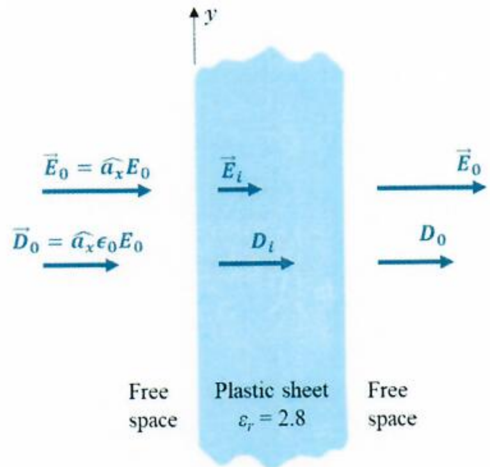


Fig. Q1. A plastic sheet in a uniform electric field intensity

(a) **[3 pts]** Determine $\overrightarrow{\boldsymbol{D_i}}$ inside the plastic sheet.
(b) **[7 pts]** Determine $\overrightarrow{\boldsymbol{E_i}}$ inside the plastic sheet.
(c) **[10 pts]** Find $\overrightarrow{\boldsymbol{P_i}}$, and $\overrightarrow{\boldsymbol{P_0}}$ inside, and outside the plastic sheet, respectively.
(d) **[5 pts]** Find the polarization induced charge density in the plastic sheet.

We can assume that the introduction of the plastic sheet ($\epsilon_r = 2.8$) does not disturb the original $\vec{E}_0 = \hat{a}_x E_0$. Additionally, the $\vec{E}$ perpendiculaly propagate to the interface, thus x-diretion is normal direction.

$$E_x = E_n$$

And, there is no free charges at the interface.

(a) $\vec{D}_i = \hat{a}_x D_i = \hat{a}_x D_0$ (D_n is continuos across the boundary)

So, $\vec{D}_i = \epsilon_0 E_0 \hat{a}_x$

(b) In the plastic sheet,

$$\vec{D}_i = \epsilon_0 \epsilon_r \vec{E}_i \rightarrow \vec{E}_i = \frac{1}{\epsilon_0 \epsilon_r} \vec{D}_i$$

$$\vec{E}_i = \frac{1}{\cancel{\epsilon_0}\epsilon_r}(\cancel{\epsilon_0} E_0 \hat{a}_x) = \hat{a}_x \frac{E_0}{\epsilon_r} = \hat{a}_x \frac{E_0}{2.8}$$

(c) To first find $\vec{P_i}$, $\vec{D_i} = \epsilon_0 \vec{E_i} + \vec{P_i} \rightarrow \vec{P_i} = \vec{D_i} - \epsilon_0 \vec{E_i}$

Therefore, from parts (a) & (b)

$$\vec{P_i} = \hat{a}_x \epsilon_0 E_0 - \epsilon_0 \hat{a}_x \frac{E_0}{2.8}$$

$$= \hat{a}_x \epsilon_0 E_0 \left(1 - \frac{1}{2.8}\right)$$

$$= \hat{a}_x 0.643 \epsilon_0 E_0$$

In addition, the polarization vector in outside the plastic sheet is 0, due to no dielectric materials, so

$$\vec{P_o} = 0$$

(d) In the plastic sheet, the polarization induced charge, ρ_p due to ϵ_r will be;

$$\rho_p = -\nabla \cdot \vec{P} = -\left(\frac{\partial}{\partial x}\hat{a}_x + \frac{\partial}{\partial y}\hat{a}_y + \frac{\partial}{\partial z}\hat{a}_z\right) \cdot (0.643 \epsilon_0 E_0 \hat{a}_x)$$

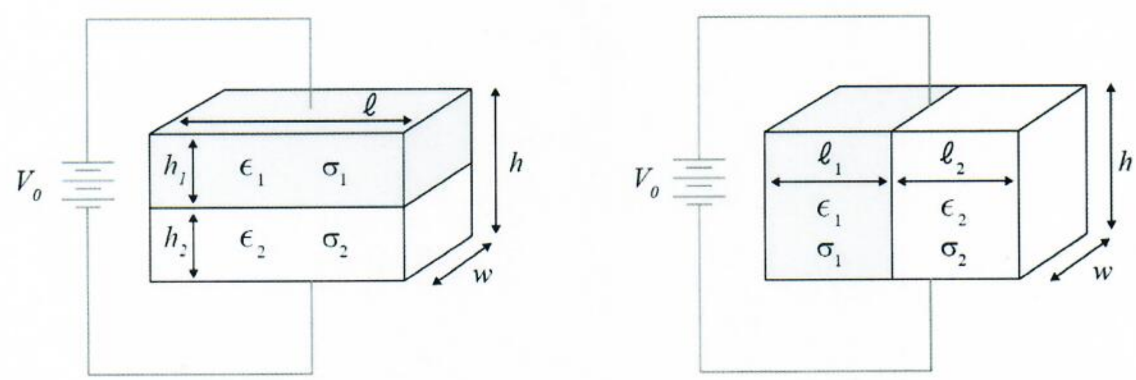
$$= 0$$

Therefore,

$$\rho_p = 0$$

Q2 **[25 points]** The two capacitors below consist of two conducting plates with two dielectric materials between them on the arrangements shown. Assume $\boldsymbol{h \ll l}$ and $\boldsymbol{h \ll w}$. Note that $\boldsymbol{h = h_1 + h_2}$, and $\boldsymbol{l = l_1 + l_2}$.

a) **[4 pts]** Determine the total capacitance in both Case 1 and Case 2 if $\boldsymbol{\epsilon_1 = \epsilon_2}$
b) **[4 pts]** Determine the total leakage resistance in both Case 1 and Case 2.
c) **[6 pts]** Determine the total capacitance in Case 1 if $\boldsymbol{\epsilon_1 = 4\epsilon_2}$, and $\boldsymbol{h_1 = 1/2h_2}$
d) **[6 pts]** Determined the total leakage resistance of Case 2 if $\boldsymbol{\sigma_1 = 1/3\,\sigma_2}$ and $\boldsymbol{l_1 = 2l_2}$
e) **[5 pts]** Draw the $\vec{\boldsymbol{E}}$-field and current density $\vec{\boldsymbol{J}}$ distribution in each case. Qualitatively explain your illustration.



Case 1 Case 2

Fig Q2. Two configuration of plate capacitors

The two capacitors have different connections of dielectric materials.
Case I : Two dielectric materials are connected in series
Case II : Two dielectric materials are connected in parallel.

(a) $C_1 = \epsilon_1 \frac{lw}{h_1}$, $C_2 = \epsilon_2 \frac{lw}{h_2}$ (Case I), So $C_{total} = \frac{C_1 C_2}{C_1 + C_2} = \frac{\epsilon_1 \epsilon_2 l w}{\epsilon_1 h_2 + \epsilon_2 h_1}$

where, $\epsilon_1 = \epsilon_2$, so $C_{total} = \frac{\epsilon_1^2 (lw)}{h_1 + h_2}$ or $\frac{\epsilon_2^2 (lw)}{h_1 + h_2}$

(Case II)

$C_{total} = C_1 + C_2 = \epsilon_1 \frac{l_1 w}{h_1} + \epsilon_2 \frac{l_2 w}{h} = \frac{(\epsilon_1 l_1 + \epsilon_2 l_2) w}{h}$

where, $\epsilon_1 = \epsilon_2$ $C_{total} = \frac{\epsilon_1 w (l_1 + l_2)}{h}$ or $\frac{\epsilon_2 w (l_1 + l_2)}{h}$

(b) $R = \frac{l}{\sigma S}$,

(Case I), R is in series. $R_{total} = R_1 + R_2 = \frac{h_1}{\sigma_1(lw)} + \frac{h_2}{\sigma_2(lw)}$

$$= \frac{\sigma_2 h_1 + \sigma_1 h_2}{\sigma_1 \sigma_2 (lw)}$$

(case II), R is in parallel, $R_{total} = \frac{R_1 R_2}{R_1 + R_2}$, where $R_1 = \frac{h}{\sigma_1(l_1 w)}$, $R_2 = \frac{h}{\sigma_2(l_2 w)}$

Therefore, $$R_{total} = \frac{\frac{h^2}{\sigma_1 \sigma_2 l_1 l_2 w^2}}{\frac{h}{\sigma_1(l_1 w)} + \frac{h}{\sigma_2(l_2 w)}} = \frac{h}{w(\sigma_2 l_2 + \sigma_1 l_1)}$$

(c) In part (a) (Case I)

$C_{total} = \frac{\epsilon_1 \epsilon_2 (lw)}{\epsilon_1 h_2 + \epsilon_2 h_1}$, $\epsilon_1 = 4\epsilon_2$, $h_1 = \frac{1}{2} h_2$ ($\to h_2 = 2h_1$)

Therefore, $C_{total} = \frac{4\epsilon_2^2 (lw)}{8\epsilon_2 h_1 + \epsilon_2 h_1} = \frac{4}{9}\left[\frac{\epsilon_2 (lw)}{h_1}\right]$

or, writing in terms of ϵ_1 and h_2

$$C_{total} = \frac{\frac{1}{4}\epsilon_1^2 (lw)}{\epsilon_1 h_2 + \frac{1}{8}\epsilon_1 h_2} = \frac{2}{9}\left[\frac{\epsilon_1 (lw)}{h_2}\right]$$

(d) In part (b), Case II

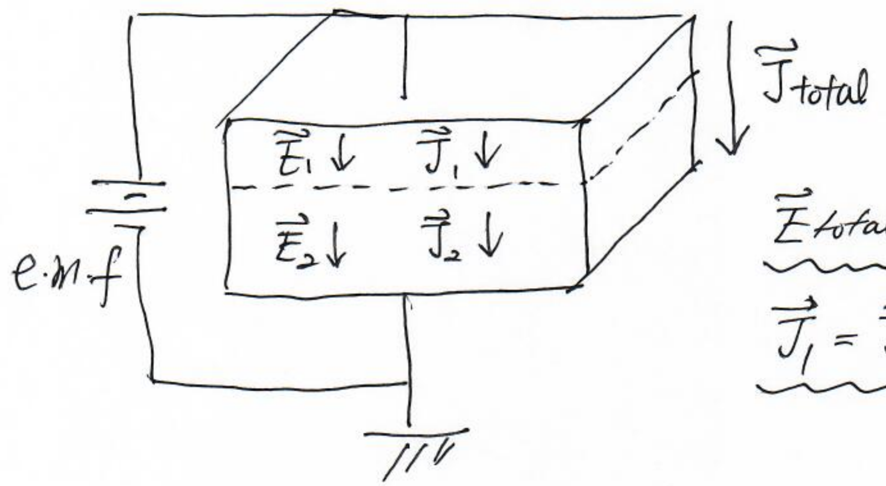
$R_{total} = \frac{h}{w(\sigma_2 l_2 + \sigma_1 l_1)}$, if $\sigma_1 = \frac{1}{3}\sigma_2$ and $l_1 = 2 l_2$

$$R_{total} = \frac{2}{5}\left(\frac{h}{w}\right)\left(\frac{1}{\sigma_1 l_1}\right)$$

or if $\sigma_2 = 2\sigma_1$ and $l_1 = 2 l_2$

$$R_{total} = \frac{3}{5}\left(\frac{h}{w}\right)\left(\frac{1}{\sigma_2 l_2}\right)$$

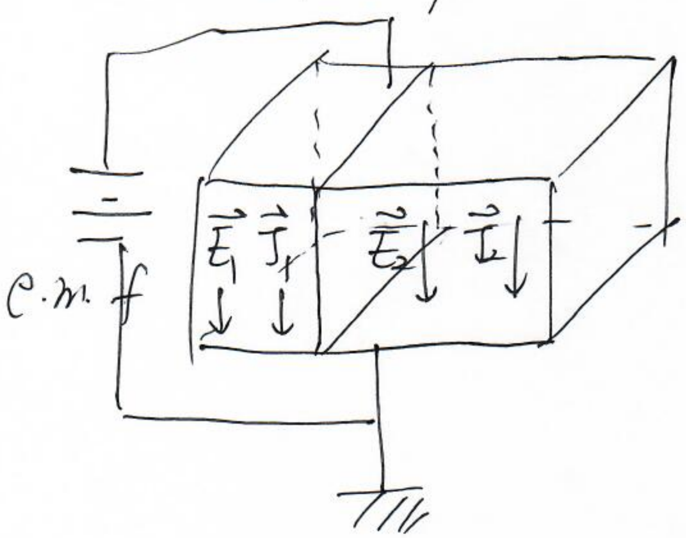
(e) Case I (Capacitor in series)



$\vec{E}_{total} = \vec{E}_1 + \vec{E}_2$

$\vec{J}_1 = \vec{J}_2 = \vec{J}_{total}$

Case II, (Capacitor in parallel)



$\vec{E}_{total} = \vec{E}_1 = \vec{E}_2$

$\vec{J}_{total} = \vec{J}_1 + \vec{J}_2$

Q3 **[25 points]** Two finite insulated conducting planes maintained at potentials **0** and $\boldsymbol{V_0}$ form a wedge-shaped configuration, as shown in Figure Q3.

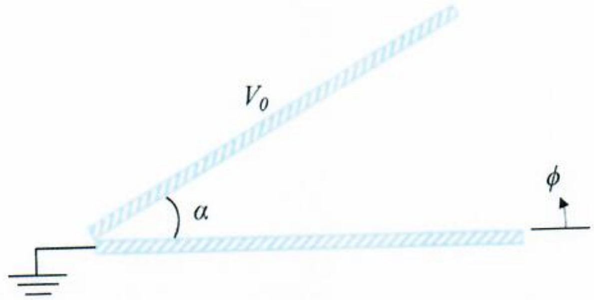


Fig. Q3

(a) **[10 pts]** Determine the **potential distributions** and the $\vec{E}$ at $\mathbf{0 < \phi < \alpha}$
(b) **[15 pts]** Determine the **potential distributions** and the $\vec{E}$ at $\mathbf{0 < \phi < 2\pi}$

Since the finite insulated conducting planes produce the electric potential (V) as a only function of Angle ϕ, thus Based on cylindrical coordinates, we can use Laplace's Equation to find, V,

$$\nabla^2 V = \cancel{\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial V}{\partial r}\right)}^{0} + \frac{1}{r^2}\frac{\partial^2 V}{\partial \phi^2} + \cancel{\frac{\partial^2 V}{\partial z^2}}^{0} = 0$$

Since, V is a function of ϕ, so

$$\frac{1}{r^2}\frac{\partial^2 V}{\partial \phi^2} = 0 \rightarrow \frac{\partial^2 V}{\partial \phi^2} = 0$$

$$V(\phi) = A\phi + B$$

using boundary conditions, we can determine A & B.

(a) For $0 \le \phi \le \alpha$

At $\alpha = 0$, $V(0) = 0$, so $B = 0$
At $\phi = \alpha$, $V(\alpha = \phi) = A\alpha = V_0$, so $A = V_0/\alpha$.

Therefore, $V(\phi) = \frac{V_0}{\alpha}\phi \quad 0 \le \phi \le \alpha$

$$\vec{E}(\phi) = -\nabla V(\phi) = -\frac{V_0}{\alpha}\hat{a}_\phi$$

(b) For $\alpha \leq \phi \leq 2\pi$;

At $\phi = \alpha$, $V(\alpha) = A\alpha + B = V_o$

At $\phi = 2\pi$, $V(2\pi) = A2\pi + B = 0$

Solving for A, using two simultaneous Equations,

$$A = \frac{-V_o}{2\pi - \alpha}, \quad B = \frac{2\pi V_o}{2\pi - \alpha}$$

Therefore,

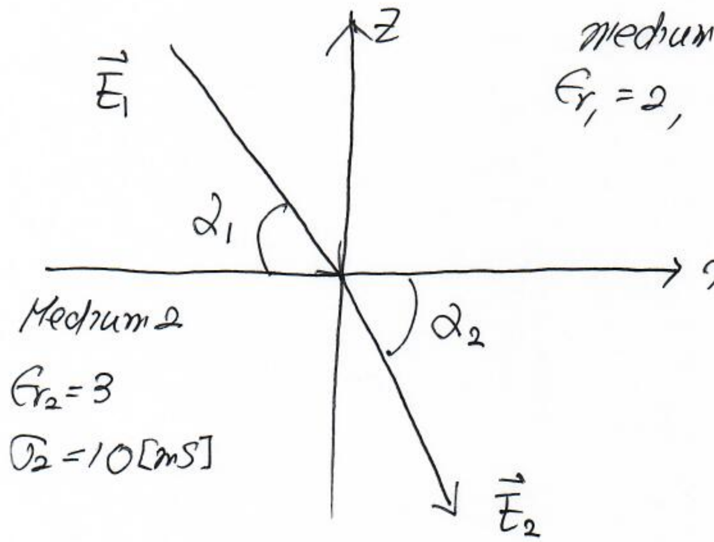
$$V(\phi) = \frac{V_o}{2\pi - \alpha}(2\pi - \phi) \quad (\alpha \leq \phi \leq 2\pi)$$

$$\vec{E}(\phi) = +\left(\frac{V_o}{2\pi - \alpha}\right)\hat{a}_\phi \quad (\alpha \leq \phi \leq 2\pi)$$

Q4 **[25 points]** Two lossy homogeneous dielectric media with dielectric constants $\epsilon_{r_1} = 2$, $\epsilon_{r_2} = 3$ and conductivities $\sigma_1 = 15\ [mS]$, $\sigma_2 = 10\ [mS]$ are in contact at the $z = 0$ plane. In the $z > 0$ region (medium 1) a uniform electric field $\overrightarrow{E_1} = \widehat{a_x}20 - \widehat{a_z}50\ [V/m]$ exists. Find;

(a) **[8 pts]** $\overrightarrow{E_2}$ in medium 2.
(b) **[6 pts]** $\overrightarrow{J_1}$ and $\overrightarrow{J_2}$.
(c) **[4 pts]** the angle $\overrightarrow{J_1}$ and $\overrightarrow{J_2}$ make with the $z = 0$ plane.
(d) **[7 pts]** the surface charge density at the interface.

From the problem conditions,



medium 1,
$\epsilon_{r_1} = 2$, $\sigma_1 = 15\,[mS]$

① $\vec{E}_1 = \hat{a}_x 20 - \hat{a}_z 50\ [V/m]$
② x-axis is tangential component
③ z-axis is normal direction

Therefore, using boundary conditions,

(a) To determine $\vec{E}_2$, from $\vec{E}_1 = \hat{a}_x 20 - \hat{a}_z 50$, $\underbrace{\vec{E}_2 = A\hat{a}_x + B\hat{a}_z = A\hat{a}_t + B\hat{a}_n}$

① $E_{2t} = E_{1t} = E_{2x} = E_{1x}$

$E_{2t} = E_{1t} = \underline{20}$

② $\vec{J}_1 = \sigma_1 \vec{E}_1$, $\vec{J}_2 = \sigma_2 \vec{E}_2$ $\langle E_{1n}, E_{2n}, E_{1x}, E_{2x} \rangle$

where, $J_{1n} = J_{2n}$, so $\sigma_1 E_{1n} = \sigma_2 E_{2n}$

Therefore, $E_{2n} = \dfrac{\sigma_1}{\sigma_2} E_{1n} = \left(\dfrac{15}{10}\right)(-50) = \underline{-75}$

$\therefore\ \vec{E}_2 = A\hat{a}_x + B\hat{a}_z = A(\hat{a}_t) + B(\hat{a}_n)$

$= \hat{a}_x 20 - \hat{a}_z 75$

(b) $\vec{J}_1 = \sigma_1 \vec{E}_1$, so $\vec{J}_1 = 15\times10^{-3}(\hat{a}_x 20 - \hat{a}_z 50)$

$= 0.3\hat{a}_x - 0.75\hat{a}_z$ [A/m²]

$\vec{J}_2 = \sigma_2 \vec{E}_2 = 10\times10^{-3}(\hat{a}_x 20 - \hat{a}_z 75) = \hat{a}_x 0.2 - \hat{a}_z 0.75$ [A/m²]

(c) $\alpha_1 = \tan^{-1}\left(\frac{z}{x}\right) = \tan^{-1}\left(\frac{50}{20}\right) \cong 68.2°$

$\alpha_2 = \tan^{-1}\left(\frac{z}{x}\right) = \tan^{-1}\left(\frac{75}{20}\right) \cong 75.1°$

(d) Using boundary condition,

$D_{2n} - D_{1n} = \epsilon_2 E_{2n} - \epsilon_1 E_{1n} = \rho_s$ (surface charge density)

where, $E_{1n} = 20$, $E_{2n} = -75$. & $E_n = \frac{\rho_s}{\epsilon_0}$

$\rho_s = \epsilon_0 (\epsilon_2 E_{2n} - \epsilon_1 E_{1n})$

$= \epsilon_0 (-3\times75 + 2\times50) = -125\epsilon_0$ [C/m²]

or if you use $\epsilon_0 = \frac{1}{36\pi}\times10^{-9}$

Then also

$\rho_s = -125\times\frac{1}{36\pi}\times10^{-9} = -1.105$ [nC/m²]

THE END